Remagnetization in Keweenawan rocks. Part II: lava flows within the Copper Harbor Conglomerate, Michigan

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Alternating field and thermal demagnetization studies of a thin lava flow unit within the Copper Harbor Conglomerate, Michigan Native Copper District, show a westerly increase in chemical remanent magnetization (CRM) overprinting along a 50 km strike length. The CRM may have formed while secondary minerals, including native copper, were being generated in the underlying Portage Lake volcanics, and when a rhyolite plug was injected into nearby siltstones of the overlying Freda Formation.

The direction of magnetization during Copper Harbor time as given by the flows is virtually indistinguishable from that found from the Portage Lake volcanics, thus supporting previous geological observations that the two formations are closely associated in time.

Des études de désaimantation thermique et par champ alternatif d'une unité mince d'une coulée volcanique à l'intérieur du conglomérat Copper Harbor, district du Michigan à cuivre natif, montrent un accroissement vers l'ouest en superimpression de l'aimantation rémanente chimique (ARC) dans une direction de 50 km de longueur. L'ARC s'est développée lorsque les minéraux secondaires, incluant le cuivre natif, était en voie de formation dans les roches volcaniques sous-jacentes du lac Portage, et lorsque le bouchon de rhyolite fut injecté dans les siltstones avoisinants la formation Freda sous-jacente.

La direction d'aimantation durant le temps Copper Harbor telle qu'indiquée est pratiquement indiscernable de celle observée à partir de roches volcaniques du lac Portage, en accord avec les observations géologiques antérieures associant les deux formations de très près dans le temps.

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Introduction

In a previous paper (Palmer *et al.* 1981) it was demonstrated from a paleomagnetic study of 150 basaltic pebbles that the Late Precambrian Copper Harbor Conglomerate on the Keweenaw Peninsula, Michigan had suffered a pre-folding, partial remagnetization. This overprint, shown to be a chemical remanent magnetization (CRM), was found to increase westwards along strike in apparent sympathy with an increase in burial metamorphism in the underlying Portage Lake Volcanic Group, as interpreted from secondary mineral zonation (Stoiber and Davidson 1959).

In this paper we present paleomagnetic results from thin lava flow units within the conglomerate, spanning a total strike length of more than 60 km (Fig. 1). The purpose of this investigation is threefold: (i) to resolve an uncertainty concerning the direction of the Earth's magnetic field during Copper Harbor time; (ii) to document more closely the reality of the westward increase in Copper Harbor overprinting, by examining changes in magnetization characteristics along strike within a single stratigraphic horizon; and (iii) to assess whether the overprinting has affected underlying Portage Lake volcanics, and whether it could be genetically related to native copper deposits found in these rocks.

General geology

Stratigraphy

The Copper Harbor Conglomerate is a poorly stratified boulder to pebble conglomerate with interbedded lava flows and lenses of sandstone. On the Keweenaw Peninsula (Fig. 1) the conglomerate (about 1600 m thick) is sandwiched between, and locally interfingers with, underlying mafic extrusives of the Portage Lake Group (at least 5000 m thick) and overlying shales, sandstones, and minor conglomerate of the Nonesuch and younger Freda Formations (combined thickness more than 4000 m). The entire sequence is stratigraphically conformable, and represents the type section for Late Precambrian (~1.1 Ga old) Keweenawan rocks of the Lake Superior region.

Within the conglomerate detailed mapping (White *et al.* 1953; Cornwall and Wright 1954; Cornwall 1955) has revealed three lava flow units: (a) a lower, unexposed one of perhaps two flows that on drill hole and magnetic anomaly evidence is present only in the

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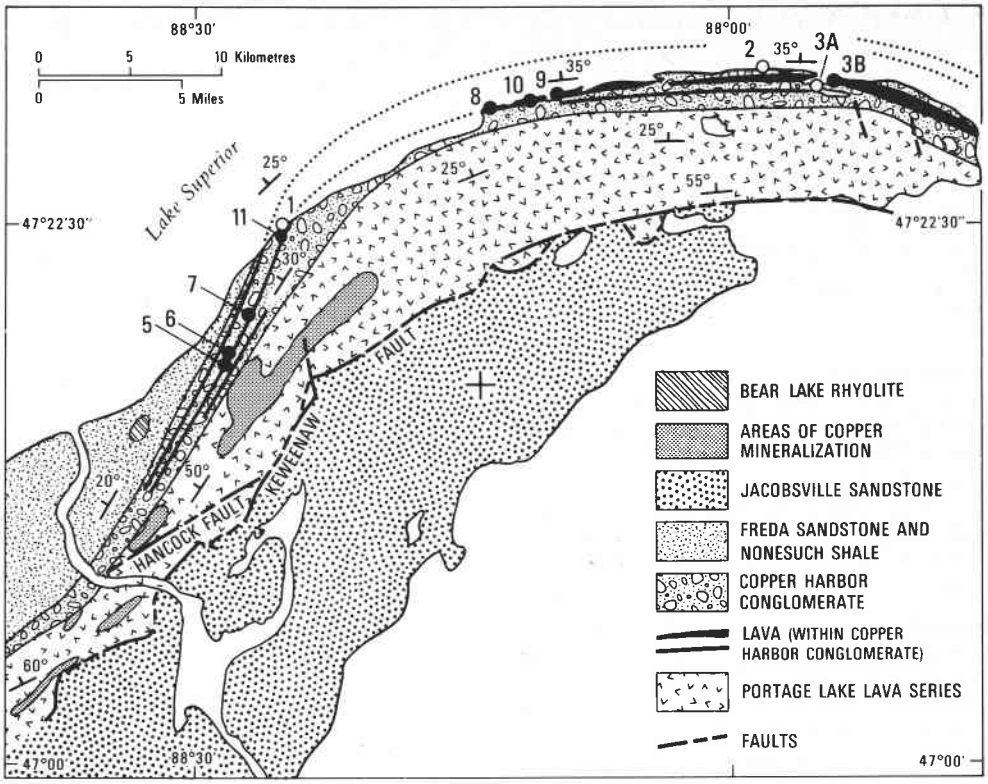


FIG. 1. Geological map of the Keweenaw Peninsula showing location of sampling sites within the Copper Harbor Conglomerate. Open dots refer to sites 1–3A where lava pebbles only were collected (see Palmer *et al.* 1981). Closed dots are sites from *in situ* Copper Harbor lava units.

western half of Fig. 1; (b) an upper unit comprising 8–10 flows that is present throughout the area of Fig. 1 but can only be followed from magnetic anomalies to the east where it lies offshore beneath Lake Superior; and (c) an intermediate one of comparable lateral extent but which is of particular interest here as it can be sampled along almost its entire strike length. In the western half of Fig. 1 the intermediate unit is present as two flows, each less than 20 m thick, separated by about 30 m of conglomerate. The flow sequence thickens eastward until it comprises about 15–20 flows with a total thickness of about 400 m at the most easterly sampled locality (Fig. 1).

West of the Keweenaw Peninsula, in the Porcupine Mountains (120 km west of Fig. 1), the Copper Harbor Conglomerate thins to about 160 m where it overlies a local pile of felsic and andesitic flows, about 2500 m thick, referred to provisionally as the "unnamed" formation by White (1972). This extrusive sequence overlies the more basaltic flows of the Portage Lake Group and may thus, at least in part, be contemporaneous with the lower part of the Copper Harbor Conglomerate on the Keweenaw Peninsula (Hubbard 1975; White 1972).

Structure

The entire Keweenawan sequence along the Keweenaw Peninsula is part of a broad northward-plunging anticline that forms a segment of the southern limb of the Keweenawan Lake Superior basin (e.g., Halls 1966). The base of the sequence is not seen, as the Keweenaw fault (Fig. 1) has downthrown a formation known as the Jacobsville Sandstone on its south side against the Portage Lake volcanics. The Jacobsville is sharply upturned against the fault and has also been involved, along with Paleozoic rocks, in faulting at Limestone Mountain (about 20 km to the southwest of Fig. 1). Apart from these tectonic regions, the Jacobsville is flat lying. It is known to underlie Upper Cambrian rocks and on indirect evidence is thought to postdate the Freda Sandstone and thus to represent a later phase of Keweenawan basin infilling (e.g., Hamblin 1961; Halls and West 1971).

In general, the dip of the Portage Lake volcanics decreases northwards away from the Keweenaw fault from a maximum 60° to about 30°. Dips of less than 10° are recorded from the Nonesuch and Freda Formations at larger distances from the fault. No more than a third of

the presently observed dip at any one locality north of the fault can be attributed to basinal subsidence during volcanism and subsequent sedimentation (White 1971). Further steepening occurred in post-Freda time. If movement on the Keweenaw fault also occurred at this time, then the tectonism is post-Jacobsville, and may even in part be post-Devonian if the Limestone Mountain disturbance is a related event.

Metamorphism and native copper mineralization

The Keweenaw Peninsula has long been famous for native copper mineralization that occurs extensively in the Portage Lake volcanics (Butler and Burbank 1929). The copper tends to be associated with other secondary minerals such as chlorite and epidote, whose formation has been attributed to burial metamorphism (Stoiber and Davidson 1959; Jolly and Smith 1972). These secondary minerals, like the copper, are most abundant in flow tops and interflow conglomerates, in contrast to the more massive less permeable flow interiors, which are relatively fresh. In general the secondary mineralization occurs as fillings in amygdules and interfragmental spaces within flow tops and conglomerates, but may also replace the host rock (White 1971).

The level of burial metamorphism, defined by mineral zonation, increases with depth in the lava pile but also increases toward the west (Stoiber and Davidson 1959). The highest level attained is defined by the appearance of epidote and pumpellyite, suggesting that maximum temperatures reached were at least 300°C.

The distribution of copper appears to parallel mineral zonation, and is closely associated with lower grade secondary minerals such as quartz, datolite, and prehnite. It appears to have formed later than most chlorite, epidote, and pumpellyite but earlier than zeolites (Butler and Burbank 1929; White 1971). According to White (1971) "there is nothing to suggest an important break in a rather continuous sequence of mineralisation governed by falling temperature."

Of particular relevance to the present study is that the vast majority of native copper deposits occur in the western part of the Keweenaw Peninsula (Fig. 1). This localization, which is also accompanied by a relatively high level of burial metamorphism compared with that experienced by similar stratigraphic horizons further east, may be a manifestation of two events. The first is that the copper occurs in an area close to a possible migration route of ore-bearing fluids ascending the steepest parts of the limbs of the Keweenawan basin (White 1971) and the second, possibly related, is that the copper occurs immediately updip from the only known post-Freda intrusion, the Bear Lake Rhyolite (Fig. 1).

The exact time of copper emplacement is problematical. The close association of copper with secondary mineral zones implies that it was introduced during burial metamorphism, presumably after deposition of the Freda and possibly after Jacobsville time when the Lake Superior basin had been filled. That some deformation preceded ore emplacement is indicated by termination of ore lenses against joints, mineralized fissures cutting hundreds of metres of lavas, and a pattern of mineralization related to the geometry of gentle, minor folds (White 1971). It is not clear, however, whether this deformation is related only to the subsidence of the main basin during Keweenawan time, or whether it includes effects of the subsequently more severe tectonism that led to the disruption of the Jacobsville along the Keweenaw fault, the overturned folds of the Porcupine Mountain area, and the vertical dips observed further west in the vicinity of the Michigan–Wisconsin border. We can only say that copper mineralization postdated *some* deformation. How much is a question that paleomagnetism may ultimately be able to answer.

Previous paleomagnetic work

Initial studies on Copper Harbor lava flows and sediments by DuBois (1962) yielded a remanence direction ($D = 290°$, $I = 32°$) that is very similar to the mean value for the underlying Portage Lake volcanics (Books 1972).

Subsequently a detailed thermal and alternating field (AF) demagnetization study of seven Copper Harbor flows by Vincenz and Yaskawa (1968) revealed stable end points having a much lower inclination ($D = 282°$, $I = 12°$). These authors inferred that the higher inclination obtained by Dubois was because his samples, not having been demagnetized, still retained a small viscous remanent magnetization (VRM) component directed along the present Earth's field. Vincenz and Yaskawa thus concluded that the Copper Harbor Conglomerate was significantly younger than the Portage Lake Group and more closely related in time to the overlying Nonesuch and Freda Formations, which also yielded shallow inclinations (DuBois 1962; Vincenz and Yaskawa 1968).

Although the shallow inclination of the Freda–Nonesuch has now been confirmed by more extensive study (Henry *et al.* 1977), the direction of the Earth's field during Copper Harbor time remains uncertain, principally for two reasons:

(a) A positive baked contact test of a Copper Harbor lava flow at site 3B in Fig. 1 yields a typical Portage Lake direction (Palmer *et al.* 1981). A similar direction was also obtained from one of three flow sites by Books (unpublished data cited in White 1972).

(b) The unnamed extrusive sequence in the Porcupine Mountain area, which may be partly correlative with the

Copper Harbor Conglomerate, also yields a paleomagnetic direction ($D = 287°$, $I = 28°$) similar to that of the Portage Lake volcanics (Books 1972, Table 5).

Experimental procedure

At least five independently oriented samples were collected at each of eight sites using a portable field drill (see Fig. 1 for site locations). Orientations were made using both sun and magnetic compasses. Laboratory measurement procedure was the same as that adopted by Palmer *et al.* (1981) with AF demagnetization measurements being divided between a Schonstedt single-axis demagnetizer and shielded DIGICO spinner magnetometer (University of Toronto (UT)) and a three-axis tumbler demagnetizer and Schonstedt spinner magnetometer (The University of Western Ontario (UWO)). In addition to AF treatment, which was made on at least one specimen from each sample core, thermal demagnetization was also performed on about half the samples, either after AF treatment, or on second undemagnetized specimens. Most thermal experiments were carried out at UWO using a Schonstedt TSD-1 unit.

Each AF demagnetization run comprised about 15–25 individual steps, successive increments being determined by the previous degree of directional movement and intensity decay, so as to maximize the smoothness and sampling regularity of the demagnetization path. For each specimen a vector diagram (Zijderveld 1967), and vector and vector difference stereoplots were generated (e.g., Halls 1979). A component search was then made using a least-squares method based on principal component analysis, recently developed by Kirschvink (1980). This method involves least-squares fitting of lines and planes to points in three-dimensional space defined by the ends of vectors obtained after each demagnetization step. Assuming that no two components are being demagnetized simultaneously in the same proportion, the orientation of a line gives the direction of any component that is, alone, being removed in the demagnetization process, whereas planes define vector trajectories generated when two components are being removed but at different rates. The orientation of the line of intersection of two planes yields the direction of components previously determined from difference vector stereoplots (Hoffman and Day 1978; Halls 1979). In the present study linear segment fits for any one specimen were only accepted provided they corresponded to segments clearly visible on the vector diagram and provided that $\theta_L \leq 5°$ where θ_L is the mean angular standard deviation (defined by Kirschvink as: $\theta_L = \tan^{-1}[(\lambda_{int} + \lambda_{min})/\lambda_{max}]^{1/2}$ where the λ's are the eigenvalues and L designates a linear fit). For more than 80% of the linear segments, the number of points fitted per line N_L was greater than 5 and θ_L was less than 3°. A similar criterion ($N_P \geq 5$ and $\theta_P \leq 2°$) was used for fitted planes but only one acceptable "Hoffman–Day" direction was found (see Appendix 1 for complete component analysis by specimen).

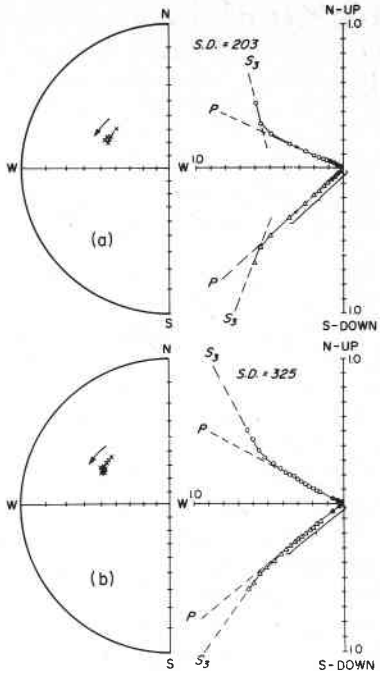


FIG. 2. Vector stereoplot and vector diagram for (*a*) specimen 302-1 from site 9 and (*b*) specimen 310-1 from site 10. For these and all subsequent plots, a structural correction has been made. In both examples the vector diagrams show (i) straight line segments decaying to the origin, corresponding to the primary or P component, and (ii) a short initial segment corresponding to a secondary component S_3, probably arising from viscous acquisition in the present Earth's field. In the vector diagrams dots represent points on the horizontal plane and triangles correspond to projections on an east–west vertical plane. Open/closed symbols refer to magnetization directions obtained after AF/thermal demagnetization. S.D. is the scale division in magnetization units (A/m $\times$ 10^{-3}). Brackets along segments indicate which points were used in linear least-squares fitting.

Experimental results

Two basic types of characteristic remanence are exhibited by Copper Harbor lavas:

(1) Single component

This type of behaviour is found only in the eastern sites 3B, 8, 9, 10, and 11 and is manifest on vector diagrams by single straight line segments that decay to the origin (Fig. 2). The direction of magnetization responsible for such segments is very similar, after structural correction, to that at site 3 where the magne-

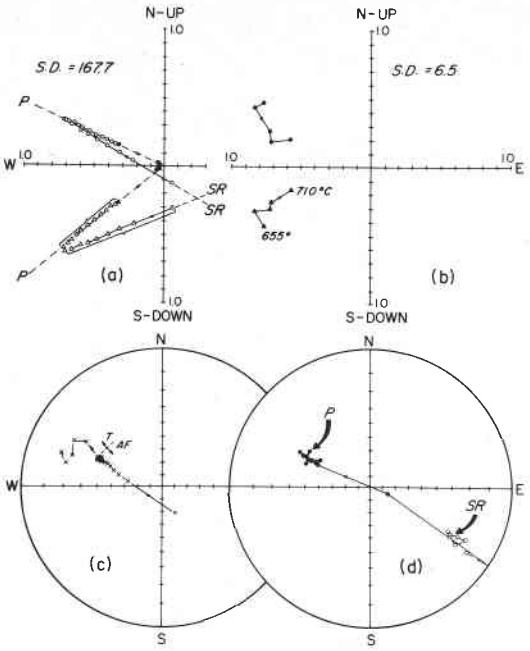


FIG. 3. Vector diagrams (*a*, *b*), vector stereoplot (*c*), and vector difference stereoplot (*d*) for specimen 299-1 from site 8. Similar to Fig. 2 except that (i) the low-coercivity component SR cannot be viscously acquired in the present field and (ii) there is a hint, revealed in the final stages of thermal treatment (diagram *b*), of secondary component(s) with inclination less than that of P. In the vector difference stereoplot (diagram *d*), open/closed symbols refer to upward/downward directed magnetizations.

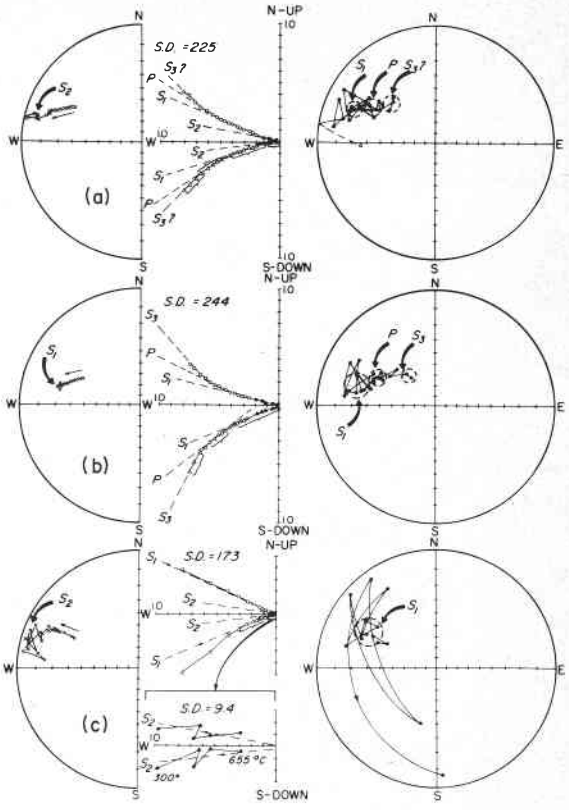


FIG. 4. Vector stereoplots, vector diagrams, and vector difference stereoplots for (*a*) specimen 277-1 from site 5, (*b*) specimen 287-1 from site 6, and (*c*) specimen 280-1 from site 5. Besides the existence of the low-coercivity component S_3 all samples exhibit clear multicomponent behaviour in contrast to specimens shown in Figs. 2 and 3. Symbols the same as in Figs. 2 and 3.

tization was shown from a positive baked contact test to be a primary remanence (P) acquired during or shortly after lava extrusion (Palmer *et al.* 1981).

Two categories of linear behaviour are shown depending upon the nature of a relatively soft component that is initially removed in alternating fields of 10 mT or less.

In Fig. 2 the soft component (S_3) is close to that of the present Earth's field (PEF) before structural correction, and is thus interpreted to be a recently acquired VRM.

In Fig. 3 the soft component has a direction that is roughly opposite to that of P. The origin of this component, designated SR, is unknown. Its low coercivity and confinement to samples that totally lack any S_3 component suggest that it is in some way locked in by internal fields generated by grains carrying the higher coercivity P remanence. The component may thus represent a partial self-reversal. Whatever its exact nature, it is, however, a real phenomenon as it is present in samples from sites 7 and 8 and in one sample from site 5 (see Appendix 2 where vector diagrams for all specimens are shown for comparison).

(2) Multicomponent

Multicomponent characteristic remanence dominates the paleomagnetic signature of western sites 5, 6, and 7 and is significant in site 11. This behaviour is revealed in vector diagrams as curved vector trajectories (Fig. 4; Appendix 2) that on close inspection are often composed of a number of linear segments, each of which is assumed to correspond to removal of a discrete remanence component. In addition to an initial low coercivity component (S_3 and in one case SR) at least three other components can be identified corresponding to P and higher coercivity secondary remanences designated S_1 and S_2 (Fig. 4). Whereas the baked contact test at site 3B and a positive fold test demonstrate that P predates deformation, the same can only be surmised for S_1 and S_2. Unlike P, which is found along the entire strike length of the lava unit, S_1 and S_2 are confined to western sites with similar structural attitude. However, if a

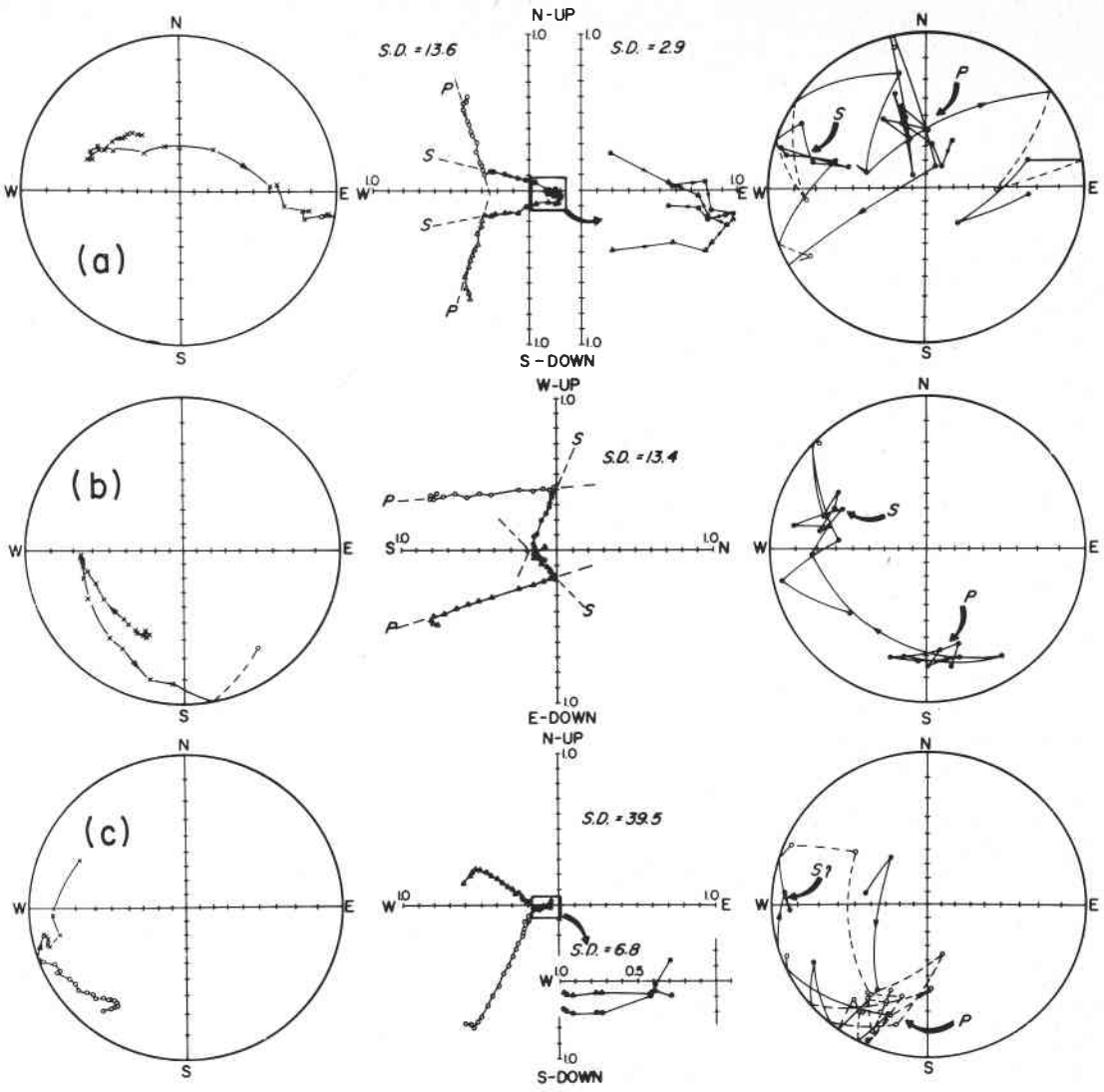


FIG. 5. Same representation as in Fig. 4 but for three conglomerate lava pebbles 160-1, 196-1, and 19-1 (*a*, *b*, and *c*, respectively). Note that the primary component P has the expected random orientation between pebbles, unlike the secondary component S. In pebbles 160-1 and 196-1, S appears to have a similar direction to S_1. It does not appear to be a composite S_1–S_2 such as in Fig. 4*a*, but instead shows a hint of another possible antiparallel component. Pebble 19-1 again shows an S component without S_1–S_2 behaviour but this time its direction is closer to S_2. Symbols as defined in Figs. 2 and 3.

bedding correction is made for S_1 and S_2 their mean direction is very similar (within a few degrees) to that of the prefolding secondary component (S) found from Copper Harbor pebbles by Palmer *et al.* (1981). Furthermore, the only secondary component found in the overlying Freda Formation also predates folding (Henry *et al.* 1977). The S component has a direction (D = 290°, I = 14°) that is intermediate between those of S_1 (D = 290°, I = 20°) and S_2 (D = 290°, I = 10°). It is possible that S is itself a composite of S_1 and S_2 since it was derived from converging remagnetization circles, a method that is relatively insensitive to resolving more than one component with a direction in the general area of convergence. Accordingly, detailed thermal demagnetization was performed on six pebbles that appeared to have been magnetically reset the most, in an effort to see whether the overprint could be resolved into S_1 and S_2 components. Although examples of overprints with typical S_1 and S_2 directions were found (Fig. 5) in no case was the overprint found to be an S_1–S_2

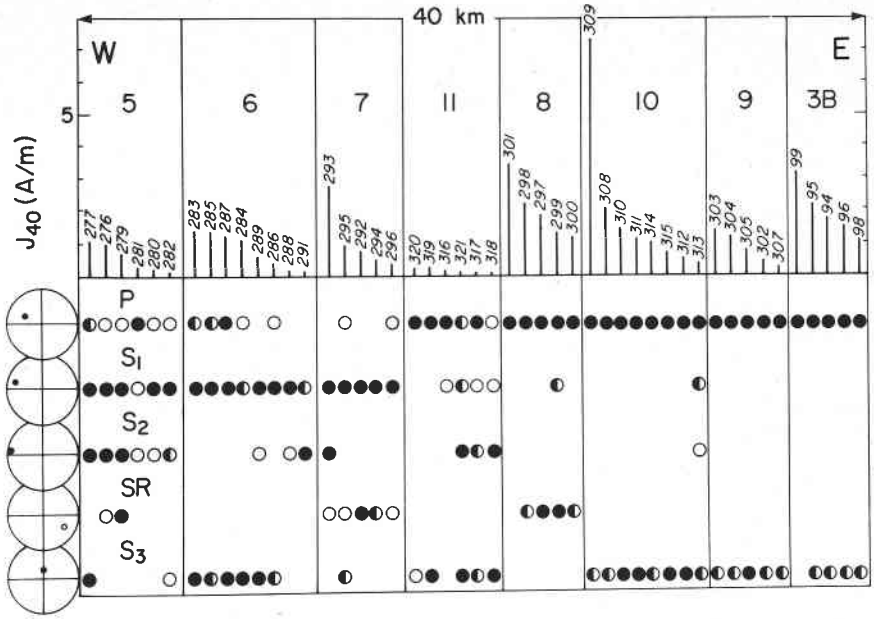


FIG. 6. Schematic representation showing the distribution of magnetization components as a function of both site and relative site location along geological strike. Solid/semi-solid/clear dots indicate that the magnetization component yields a clear linear segment/is probably present although the segment is less clear/ may be present but segments are curved or data are lacking. A blank space indicates that there is no indication, despite detailed data, that the component exists. The small stereoplots indicate the direction of each component. The intensity of each sample is also shown after a demagnetization field at 40 mT, suitable for completely removing any S_3 or SR components. Note that the westerly sites with weaker intensity show more conspicuous overprinting by S_1 and S_2 components at the expense of P. A similar but weaker relationship between intensity and overprinting is also seen at the within-site level (e.g., sites 6, 11, and 10). It is also clear that the components S_3 and SR are mutually exclusive as they are never found in the same specimen.

composite, as displayed by many of the *in situ* lava samples.

Variation of remanence characteristics along strike

A summary of paleomagnetic results is presented in Table 1 and also in Fig. 6, which shows that the mean site intensity of natural remanent magnetization (NRM) after 40 mT field treatment generally decreases westwards as the degree of magnetic overprinting increases. The S_1 and S_2 components are only positively identified west of, and including, site 11. However, hints of very minor overprinting are also seen further east at site 8 for example (Fig. 3B) where progressively lower inclinations are revealed in the last 5% of initial NRM intensity.

Discussion

The paleomagnetic data show that the direction of the primary P component in Copper Harbor lava flows is not significantly different from that in the underlying Portage Lake volcanics. This result is in harmony with the conclusion of White (1972), based on geological evidence, that temporally the Copper Harbor is more closely linked with the underlying Portage Lake lavas than the overlying Nonesuch and Freda Formations. The paleomagnetic directions are also no longer in conflict with those obtained from the unnamed unit by Books (1972).

However, we have shown that the shallow inclination remanence obtained by Vincenz and Yaskawa (1968) and regarded by them as primary is, in fact, a secondary component. The reduction in NRM intensity of lava flows (Fig. 6) and the higher blocking temperatures of either S_1 and S_2 component compared to P demonstrate that the overprinting is a CRM. A westward increase in CRM overprinting within the Copper Harbor Formation found initially from the pebble data (Palmer *et al.* 1981) has thus been confirmed from study of *in situ* lava flows.

We are uncertain about the precise composition of the overprinting episode, whether it represents two discrete events, or more plausibly a continuous acquisition of CRM through a time span sufficiently long for a measurable amount of apparent polar wander to have occurred. In this context it is important to caution against the uninhibited use of linear segments on vector diagrams. Some of the linear segments satisfying the rather stringent linearity criteria we have imposed are

TABLE 1. Summary of paleomagnetic results for Copper Harbor lava flows

Site	N	n	$D°$	$I°$	k	α_{95}
			Primary (P) component			
3B	6	6	291.9	37.4	275	4.1
9	5	5	294.4	43.2	371	4.0
10	8	8	298.2	37.7	779	2.0
8	5	5	298.7	40.0	186	5.6
11	6	5	297.4	33.4	58	10.2
6	8	3	305.9	35.9	20	28.3
5	7	2	306.0	37.1	103	24.8
			Secondary (S_1) component			
8 + 10 + 11	19	3	292.5	19.9	67	15.2
7	5	5	283.0	16.4	443	3.6
6	8	8	282.8	22.8	169	4.3
5	7	5	291.3	18.7	412	3.8
			Secondary (S_2) component			
11	6	3	284.8	8.9	389	6.3
6 + 7	13	2	283.6	10.7	3226	4.4
5	7	4	285.9	8.3	395	4.6

	N_s	$D°$	$I°$	P lat. (°N)	P long. (°E)	δm	δp
			Mean site directions and pole positions				
P	7	299.0	37.9	34.9	175.6	4.6	2.7
S_1	4	287.4	19.5	19.1	175.7	6.7	3.5
S_2	3	284.7	9.3	13.4	174.0	2.6	1.3

NOTES: Collection locality (lat 47.4°N; long 88.3°W); sites are listed in order from east to west; a complete listing of specimen results is given in Appendix 1; N is the total number of samples per site; n is the number of samples per site that yielded the direction of a given component; N_s is the total number of sites per component; D, I are the declination, inclination of remanence after structural correction; k, α_{95} are the usual Fisher statistical parameters; and δm, δp are the semi-axes of the 95% confidence oval about the pole.

still nevertheless very gently curved (Fig. 4*b*; Appendix 2) and thus the possibility that the linearity results from two components decaying at almost the same rate is very real. The acceptance of lines drawn through sets of paleomagnetic data must still in the end be determined by visual inspection of vector diagrams and stereoplots.

In the present example we feel that our data are of sufficient quality to justify a separation of components. However, there is nothing in our data that would eliminate the possibility that the multicomponent signatures seen in Fig. 4 represent remanence continuously acquired during a period of measurable apparent polar wander and that the linear segments are generated by particular component coercivity or blocking temperature relations. However, there is no doubt that the single linear segment vector diagrams, so characteristic of eastern sites, are replaced by curved multicomponent ones further west (see Appendix 2). Since the degree of CRM overprinting increases westwards in sympathy with hydrothermal–burial metamorphism in the underlying Portage Lake lavas, it is plausible that the two phenomena are genetically linked. If, as seems likely, the CRM is prefolding, the period of chemical activity and attendant secondary mineral formation occurred prior to tectonism in post-Freda time. Native copper, which is a component of this secondary mineral assemblage, would thus also have been formed at this time, contrary to White's (1971) belief that copper emplacement largely postdated tectonism. It is also possible that the Bear Lake Rhyolite was intruded during the early stages of the CRM episode as its paleomagnetic direction (D = 295°, I = 22.2° after restoring Freda to horizontal, Books 1972) is close to that of the S_1 component.

One remaining aspect concerns the extent of CRM overprinting in the Portage Lake volcanics. If secondary mineral emplacement *is* contemporaneous with CRM acquisition in the Copper Harbor Formation, magnetization directions with shallow inclination (≤20°) should be found, particularly on thermal demagnetization of Portage Lake rocks. It is noteworthy that there is no hint of a shallow overprint in Books' paleomagnetic sites from the Kearsarge area (Books 1972, his Table 2). However, all these results are quoted after a demagnetization field of only 10 mT, at which stage the magnetization is still dominated by the primary component (see, for example, the Copper Harbor flows in Fig. 3). The only published thermal treatment of Portage Lake rocks is that by Vincenz (1968) and it is interesting that the only sample showing a shallow inclination (~+5°) is one from the Kearsarge area. There is thus some indication that detailed thermal treatment on Portage Lake rocks may reveal the same CRM event as found in the Copper Harbor Formation.

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Appendix 1

Component analysis (all but S_3 structurally corrected) based on the method of Kirschvink (1980) for all Copper Harbor volcanic specimens (see Table A1). Results to be compared with specimen vector diagrams in Appendix 2. Cautionary note: The analysis of multicomponent remanence through use of linear segments may lead to misleading or incorrect results. Components are accepted according to certain ad hoc but hopefully stringent criteria, the main element being that the segment should be clearly visible on the vector diagrams. However, the identification and correlation of components between specimens and sites are based on the a priori assumption that their directions are the same. The dispersion encountered in more conventional paleomagnetic analyses is such that variations of 10–20° in direction between individual specimens might be encountered. In the present analysis where the P, S_1, and S_2 components have rather similar directions, such differences would not lead to the conclusion that the same component was common between specimens and hence there is a built-in bias towards lower dispersion values for any given component. The question concerning the reality of a component is based on the assumption that it is signified by a linear segment. In actual fact a linear segment may represent a transition between two others and a third component of intermediate direction would thus be erroneously obtained. It is possible, for example, that the P component in specimens 283, 285, and 287 at site 6 and specimen 277 at site 5 may merely represent a transition between S_3 and S_1 (see Fig. 6). It should be emphasized that even with the excellent quality of the present data such uncertainties still exist.

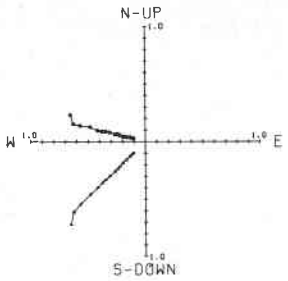
For three specimens (317–319) there is also some ambiguity regarding the direction of the P component because two possible directions were found. The more resistant one with lower inclination was chosen as it is present in all but one of the specimens from site 11. The nature of the steeper component is unknown but could represent an overlap in coercivity spectra between the P and S_3 components.

For the entire specimen suite only one acceptable Hoffman–Day direction was generated (specimen 321: $D = 301.8, I = -4.2, \overline{N}_P = 5.5, \theta_P = 1.8, 0.8$).

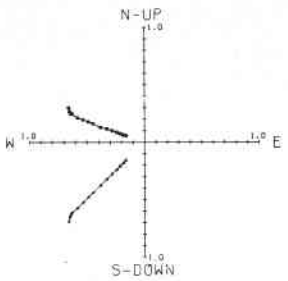
TABLE A1. Component analysis for Copper Harbor volcanic specimens

	P				S_1				S_2				S_3			
Specimen	D	I	N_L	θ_L	D	I	N_L	θ_L	D	I	N_L	θ_L	D	I	N_L	θ_L
Site 3B (bedding: strike 275°/dip 35°N)																
94	289.9	41.5	9	1.6										*		
95	289.8	42.7	12	0.7										*		
96	293.2	36.6	8	1.1										*		
97	297.3	34.8	7	2.4										*		
98	293.1	36.6	8	1.0										*		
99	288.0	31.9	11	3.2												
Site 9 (260°/30°N)																
302	294.8	40.8	16	1.0										*		
303	299.2	45.5	8	1.8										*		
304	294.9	44.6	19	1.2										*		
305	296.7	40.2	15	3.4									11.0	74.6	3	2.2
307	286.5	44.4	9	4.4										*		
Site 10 (255°/32°N)																
308	301.2	39.5	9	1.1										*		
309	296.5	36.0	7	2.1										*		
310	299.2	35.6	20	1.3									320.1	63.4	3	1.6
311	296.1	38.7	16	1.8									338.7	74.1	3	1.5
312	293.7	35.9	12	1.7									313.5	67.2	3	1.2
313	299.9	38.5	15	1.8	287.0	22.2	5	3.6						*		
314	296.4	39.2	5	1.9										*		
315	302.9	37.6	13	2.6										*		
Site 8 (250°/32°N)																
297	308.3	41.0	11	0.7												
298	293.9	43.1	17	1.3												
299	295.5	38.9	12	1.1	286.2	18.0	3	4.8		*						
300	299.0	33.2	17	1.8												
301	296.7	43.0	9	0.9												
Site 11 (225°/25°W)																
316	294.4	32.9	20	3.8												
317	285.2	32.2	6	5.0		*			283.6	10.3	4	2.7		*		
	(293.4	53.0	5	4.8)												
318		*				*			281.7	9.3	3	4.3	313.6	78.1	3	1.2
	(310.5	49.7	5	4.6)												
319	289.5	33.3	12	3.2									60.5	75.3	3	1.1
	(303.6	51.5	13	3.5)												
320	300.3	32.4	6	1.1										*		
321	318.0	33.5	8	5.0	304.4	19.1	6	4.9	289.0	7.0	6	5.0	345.9	71.5	5	1.3
Site 7 (200°/30°W)																
292					279.0	15.5	4	0.5								
293					288.5	17.3	12	1.5	283.0	11.5	9	2.0				
294					280.3	16.2	16	1.3								
295		*			284.7	14.7	10	1.6						*		
296		*			282.5	18.0	6	0.7		*						
Site 5 (200°/30°W)																
276		*			290.0	20.2	7	2.8	283.3	7.0	10	1.3				
277	313.0	36.8	4	1.9	290.0	15.5	7	2.4	283.3	6.6	6	2.6				
279		*			288.4	19.7	7	1.8	291.2	9.3	8	1.7				
280		*			296.5	21.9	9	2.1		*						
281	298.9	37.0	7	0.9		*				*						
282		*			291.5	16.3	8	2.7	285.7	10.2	4	3.3				
Site 6 (200°/30°W)																
283	331.9	36.9	4	1.1	283.3	21.0	6	1.2						*		
284		*			290.9	26.3	7	2.9						*		
285	291.0	32.4	4	1.0	271.1	20.8	6	1.6						*		
286		*			278.0	23.2	7	1.5						*		
287	296.5	34.2	6	1.5	281.2	21.9	5	1.8					8.6	72.3	3	1.6
288					287.7	23.4	11	1.8		*						
289					287.0	25.9	15	2.2		*				*		
291					283.5	19.4	4	0.6	284.1	9.8	3	1.0				

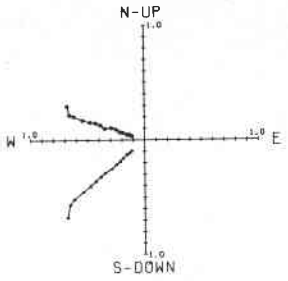
*The component is probably present but either in the case of P or S_3, no linear segments satisfying the acceptance criteria are generated, or for the more resistant components S_1 and S_2, no stable end point is reached.



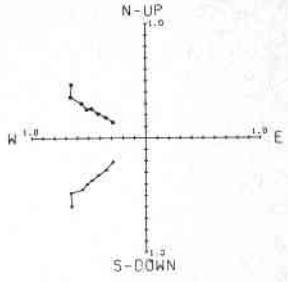
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AF: 0-100 mT



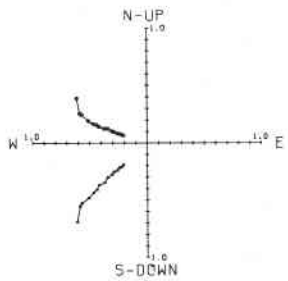
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S.D. = 393. A/m x 10^{-3}
AF: 0-100 mT



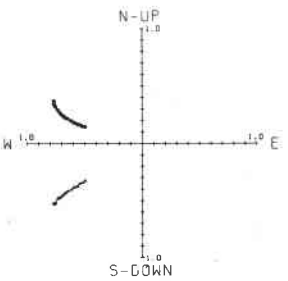
CH 96-1 N:15 SITE: 3B
S.D. = 390. A/m x 10^{-3}
AF: 0-100 mT



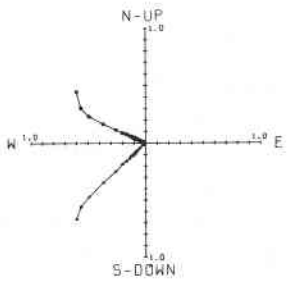
CH 97-1 N:08 SITE: 3B
S.D. = 557. A/m x 10^{-3}
AF: ZERO T:20-507° C



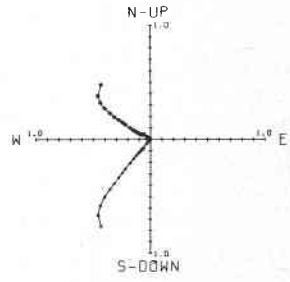
CH 98-1 N:15 SITE: 3B
S.D. = 249. A/m x 10^{-3}
AF: 0-100 mT



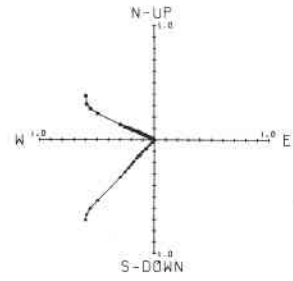
CH 99-1 N:17 SITE: 3B
S.D. = 411. A/m x 10^{-3}
AF: 0-100 mT



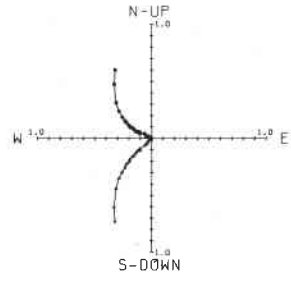
CH 302-1 N:23 SITE: 9
S.D. = 203. A/m x 10^{-3}
AF: 0-100 mT + T: 300-680° C



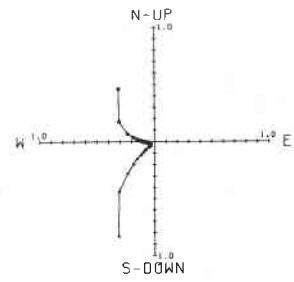
CH 303-1 N:22 SITE: 9
S.D. = 372. A/m x 10^{-3}
AF: 0-100 mT + T: 300-690° C



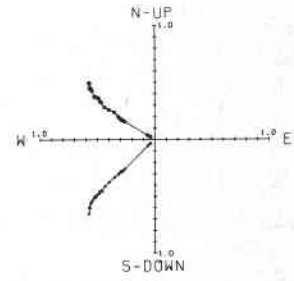
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AF: 0-80 mT + T: 300-680° C



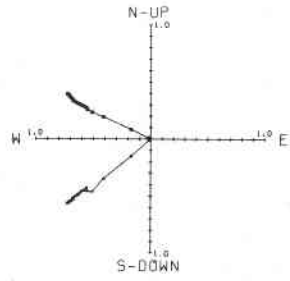
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S.D. = 274. A/m x 10^{-3}
AF: 0-100 mT + T: 300-690° C



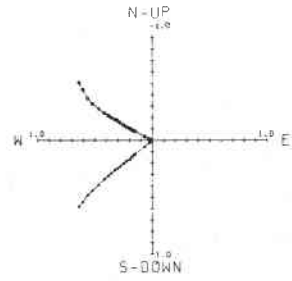
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AF: 0-100 mT



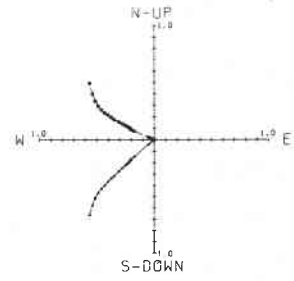
CH 308-1 N:22 SITE: 10
S.D. = 280. A/m x 10^{-3}
AF: 0-100 mT + T: 300-680° C



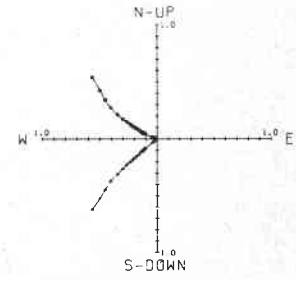
CH 309-1 N:19 SITE: 10
S.D. = 822. A/m x 10^{-3}
AF: 0-100 mT + T: 300-680° C



CH 310-1 N:23 SITE: 10
S.D. = 325. A/m x 10^{-3}
AF: 0-100 mT + T: 300-705° C

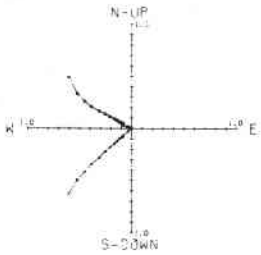


CH 311-1 N:22 SITE:10
S.D. = 244. A/m x 10^{-3}
AF: 0-100 mT + T: 300-690° C

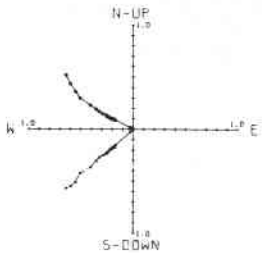


CH 312-1 N:23 SITE: 10
S.D. = 200. A/m x 10^{-3}
AF: 0-100 mT + T: 300-705° C

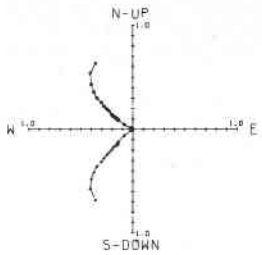
FIG. A1. Vector diagrams for Copper Harbor volcanic specimens, arranged in order of location from east to west along strike.



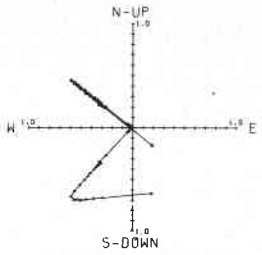
CH 313-1 N:22 SITE:10
S.D. = 265. A/m x 10^{-3}
AF: 0-100 mT + T: 300-690° C



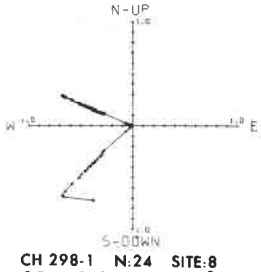
CH 314-1 N:22 SITE:10
S.D. = 297. A/m x 10^{-3}
AF: 0-100 mT + T: 300-680° C



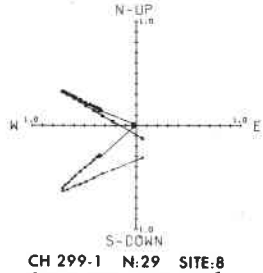
CH 315-1 N:22 SITE:10
S.D. = 223. A/m x 10^{-3}
AF: 0-100 mT + T: 300-690° C



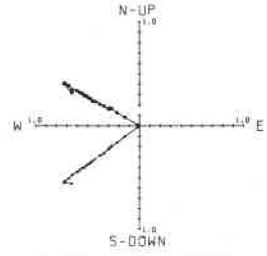
CH 297-1 N:27 SITE:8
S.D. = 290. A/m x 10^{-3}
AF: 0-100 mT + T: 300-700° C



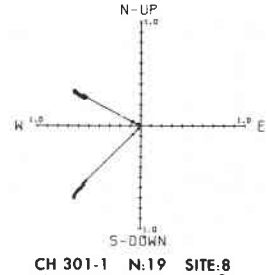
CH 298-1 N:24 SITE:8
S.D. = 363. A/m x 10^{-3}
AF: 0-100 mT + T: 300-680° C



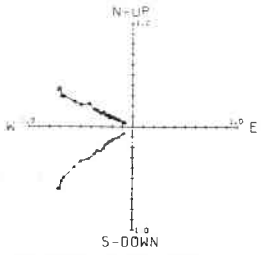
CH 299-1 N:29 SITE:8
S.D. = 168. A/m x 10^{-3}
AF: 0-100 mT + T: 300-710° C



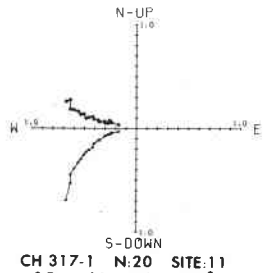
CH 300-1 N:23 SITE:8
S.D. = 189. A/m x 10^{-3}
AF: 0-100 mT + T: 300-680° C



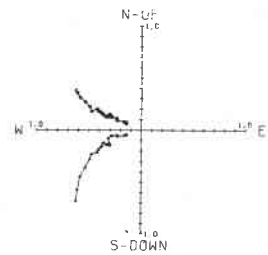
CH 301-1 N:19 SITE:8
S.D. = 384 A/m x 10^{-3}
AF: 0-100 mT + T: 300-680° C



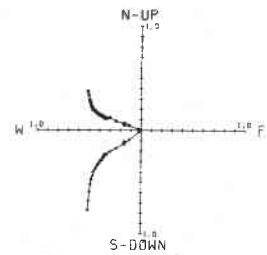
CH 316-1 N:23 SITE:11
S.D. = 21. A/m x 10^{-3}
AF: 0-100 mT + T: 300-705° C



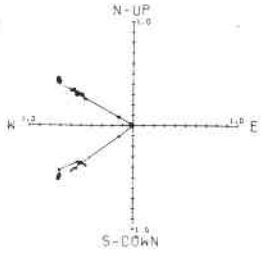
CH 317-1 N:20 SITE:11
S.D. = 11. A/m x 10^{-3}
AF: 0-100 mT + T: 300-600° C



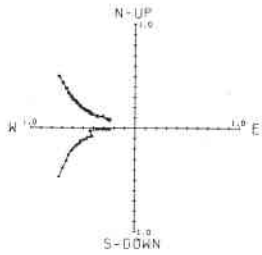
CH 318-1 N:23 SITE:11
S.D. = 5. A/m x 10^{-3}
AF: 0-100 mT + T: 300-705° C



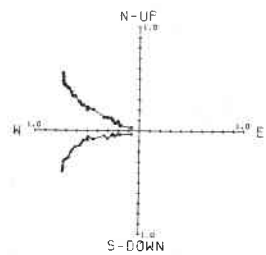
CH 319-1 N:29 SITE:11
S.D. = 47. A/m x 10^{-3}
AF: 0-100 mT + T: 300-680° C



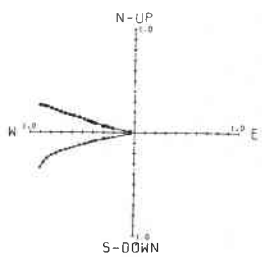
CH 320-2 N:25 SITE:11
S.D. = 29. A/m x 10^{-3}
AF: 0-100 mT + T: 300-680° C



CH 321-2 N:22 SITE:11
S.D. = 11. A/m x 10^{-3}
AF: 0-100 mT + T: 300-635° C

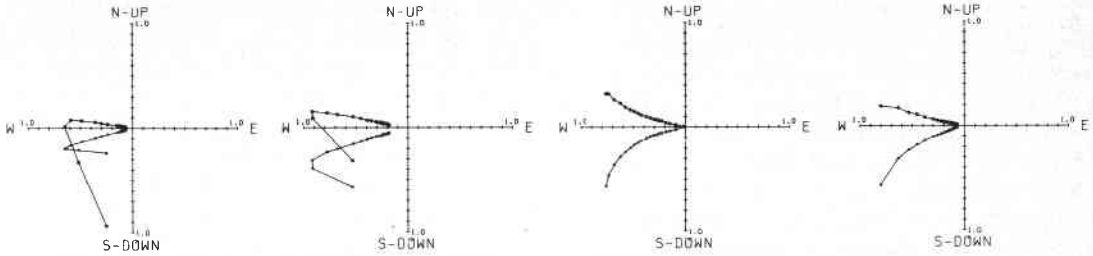


CH 321-1 N:27 SITE:11
S.D. = 13. A/m x 10^{-3}
AF: 0-100 mT + T: 400-660° C



CH 293-1 N:20 SITE:7
S.D. = 458. A/m x 10^{-3}
AF: 0-100 mT + T: 300-655° C

Fig. A1. (*Continued.*)

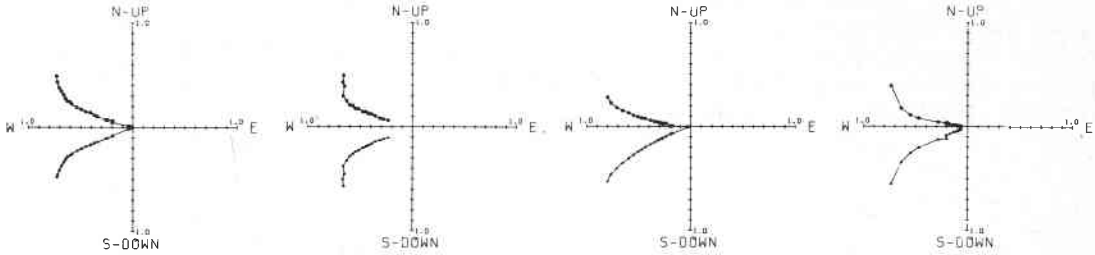


CH 292-1 N:16 SITE:7
S.D. = 484. A/m x 10^{-3}
AF: 0-100 mT

CH 294-1 N:18 SITE:7
S.D. = 171. A/m x 10^{-3}
AF: 0-100 mT

CH 295-1 N:23 SITE:7
S.D. = 212. A/m x 10^{-3}
AF: 0-100 mT + T: 300-700° C

CH 296-1 N:18 SITE:7
S.D. = 209. A/m x 10^{-3}
AF: 0-100 mT

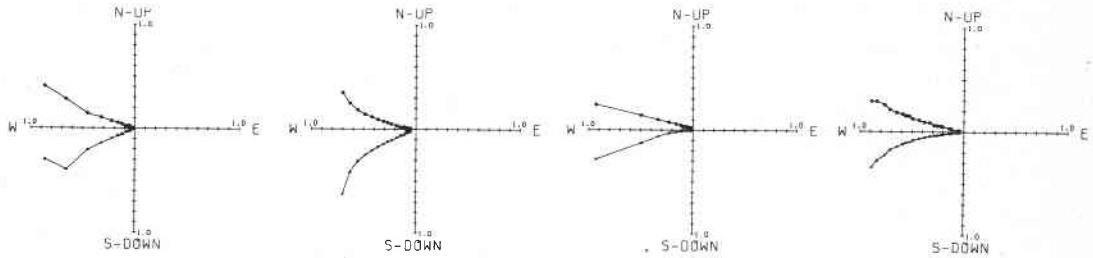


CH 283-1 N:23 SITE:6
S.D. = 212. A/m x 10^{-3}
AF: 0-100 mT + T: 300-700° C

CH 284-1 N:17 SITE:6
S.D. = 206. A/m x 10^{-3}
AF: 0-100 mT

CH 285-1 N:20 SITE:6
S.D. = 290. A/m x 10^{-3}
AF: 0-100 mT + T: 300-655° C

CH 286-1 N:14 SITE:6
S.D. = 182. A/m x 10^{-3}
AF: 0-100 mT

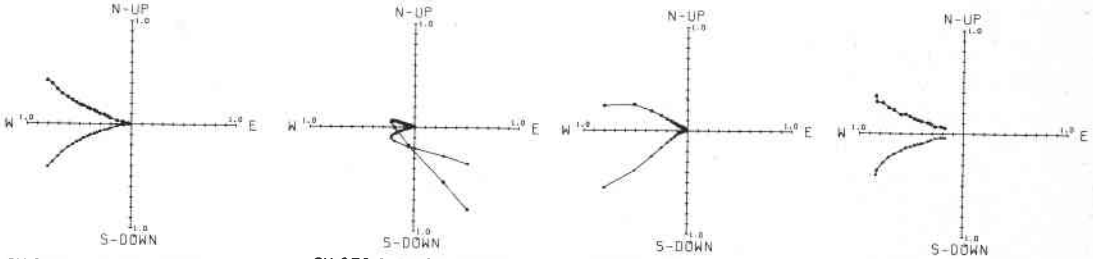


CH 288-1 N:17 SITE:6
S.D. = 133. A/m x 10^{-3}
AF: 0-100 mT

CH 289-1 N:20 SITE:6
S.D. = 234. A/m x 10^{-3}
AF: 0-100 mT + T: 300-655° C

CH 291-1 N:13 SITE:6
S.D. = 175. A/m x 10^{-3}
AF: 0-80 mT

CH 276-1 N:22 SITE:5
S.D. = 196. A/m x 10^{-3}
AF: 0-100 mT + T: 300-690° C



CH 277-1 N:22 SITE:5
S.D. = 225. A/m x 10^{-3}
AF: 0-100 mT + T: 300-655° C

CH 279-1 N:23 SITE:5
S.D. = 477. A/m x 10^{-3}
AF: 0-100 mT + T: 300-690° C

CH 281-1 N:16 SITE:5
S.D. = 233. A/m x 10^{-3}
AF: 0-100 mT

CH 282-1 N:16 SITE:5
S.D. = 22. A/m x 10^{-3}
AF: 0-100 mT

Fig. A1. (*Concluded.*)

Appendix 2

Vector diagrams (Fig. A1) for all specimens used in this study (except 287-1 and 280-1, which appear in Fig. 4) are shown here so that comparison can be made between samples carrying secondary CRM and those that do not. The specimens are listed with eastern (CRM-free) sites first and western ones last.

Beneath each vector diagram are a printout of the sample number, the number of remanence measurements used in each plot (N), the site number, the scale division (S.D.), and the demagnetization treatment (AF = alternating field; T = thermal). The interpretation of each plot, in terms of components present, is given in Appendix 1 and summarized in Fig. 6.

In general, remanence decay to the origin follows a smooth curve. However, some specimens (e.g., 309, 297–301, 318, and 321-2) show a sharp kink at the transition between AF and subsequent T demagnetization. The kink has no paleomagnetic significance as it arises from small systematic differences occasionally present between measurements made with the two spinner magnetometers in cases where the AF was performed on the DIGICO (UT) followed by thermal treatment on the Schonstedt (UWO). Although in all cases samples were re-demagnetized at the maximum AC field previously attained just prior to initial heating it is possible that small stray components acquired during the interim period were not totally removed. The maximum temperature given for each plot corresponds to the last value at which a smooth coherent directional change was observed. Heatings were performed in most cases to 700°C, but readings became erratic beyond the maximum temperature given.